A

Evidence map: what the benchmark can and cannot currently resolve

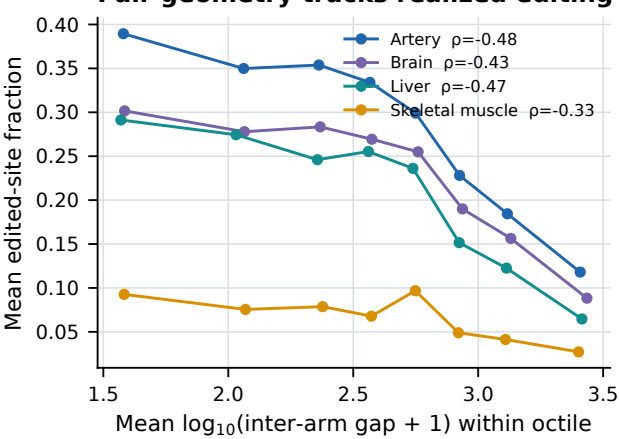
Selected · Supported · Limited · Not Supported · Deferred

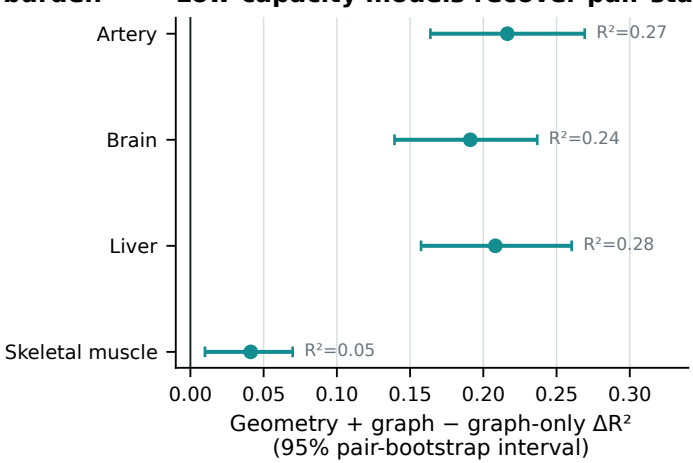
Biological / learning scale	Tested formulation	Status	Interpretation
Substrate-aware learning	Pair-first expanded supervision	Selected	Corrects label use and pair weighting
Target selectivity	Aligned target + helix-partner view	Selected	Survives shifted-window control
Integration	Fixed equal-logit global + local	Selected	Complementary same-fold predictions
Substrate realization	Pair geometry / editing burden	Supported	State is measurable, not yet actionable
Supervision	Continuous + ordinal auxiliary labels	Limited	Ranking benefit; no useful F1 gain
Target selectivity	Focal node / contact geometry	Limited	Points toward aligned local view
Cross-tissue	Shared four-tissue training	Limited	Transfer below promotion gate
Substrate realization	Hierarchical engagement gate	Limited	Small ranking gain; unstable classification
Biological context	Bulk H3K36me3 overlap	Limited	Context-specific AUPRC only
Biological context	RBP binding × bulk expression	Not Supported	No reproducible residual gain
Biological context	Transcript / repeat annotation panel	Not Supported	No robust site-level increment
Representation	Frozen RNA-language embeddings	Not Supported	No consistent incremental utility
Representation	Substrate-wide all-A decoder	Not Supported	Stopped at prespecified futility gate
Capacity	Larger / deeper graph networks	Not Supported	Capacity alone did not generalize
Integration	Unified end-to-end architecture	Not Supported	Inferior to frozen ensemble
Cellular state	Matched donor/cell multimodality	Deferred	Required measurements unavailable
RNA state	Matched in-vivo structure / isoforms	Deferred	Required measurements unavailable

Status reflects prespecified same-fold comparisons and controls; deferred hypotheses were not tested.

B

Pair geometry tracks realized editing burden



C

Low-capacity models recover pair state